

Dinesh Kannan

+1 9196335438 | Raleigh, NC | dineshkannan010@gmail.com | linkedin.com/in/dinesh012 | github.com/dineshkannan010

Education

North Carolina State University - Master of Computer Science | GPA: 3.85/4.0 **August 2023 - May 2025**

Coursework: Neural Networks & Deep Learning, Automated Learning & Data Analysis, Machine Learning with Graphs, Cloud Computing, Design & Analysis of Algorithms

Mumbai University - Bachelor of Engineering, Computer Engineering | GPA: 9.31/10 **August 2018 - May 2022**

Coursework: Machine Learning, Data Structures and Algorithms, Database Management, Object-Oriented Design

WORK EXPERIENCE

North Carolina State University | Research Assistant

July 2024 - April 2025

- Developed scalable AI Teaching Assistant chatbot using GPT-4, PyTorch, OpenAI API, and RAG pipelines, achieving 35% increase in student engagement.
- Designed RESTful APIs (Flask) optimized for high throughput, handling 1,000 concurrent queries with sub-200ms latency.
- Built full-stack web applications (React, Flask, MongoDB), improving ML response accuracy by 30%.
- Integrated automated load-testing and robust CI/CD pipelines using GitHub Actions, Docker, and AWS infrastructure.

Headstarter AI | Software Engineering Fellow

July 2024 - September 2024

- Built SaaS platform integrating OpenAI API and fine-tuned LLaMA models, enhancing contextual relevance by 33% via MongoDB vector search.
- Developed SaaS flashcard app with secure OAuth2-based login and Stripe checkout integration, reducing signup + payment flow from 1m20s to under 1m (22% faster).
- Configured CI/CD workflows with GitHub Actions and Vercel integrations, reducing deployment time by 25% by automating testing and build steps.

Deloitte | Deloitte Consulting Analyst

January 2023 - July 2023

- Software Development-** Engineered five JavaScript web API scripts for Dynamics CRM, improving data retrieval speeds by 40% and enhancing security for multiple enterprise clients.
- Automated workflows using Power BI and JavaScript, achieving a 35% efficiency improvement across teams.
- Contributed to Agile development sprints by implementing prioritized features and collaborating with cross-functional teams, helping reduce delivery time by 27%.
- Helped design and implement hybrid MySQL + MongoDB schema to support real-time analytics with reduced query latency.

Skills

AI/ML: PyTorch, TensorFlow, RAG Pipelines, Large Language Models (GPT-4, LLaMA, Claude), OpenAI API, NLP

Technical Skills: Python, JavaScript, SQL, Java, AWS Cloud, React, Next.js, Flask, Django, TypeScript, REST APIs

Tools: Git, Jira, Heroku, Power BI, Stripe, Vercel, Linux, Postman, MongoDB, AWS(EC2/S3), Google Cloud, Firebase

Projects

End-to-End Automated MLOps Pipeline for Generative AI Application

- Architected a scalable MLOps pipeline using PyTorch, Hugging Face Transformers, and AWS SageMaker, supporting GPT-4 forks.
- Developed Docker-containerized workflows with GitHub Actions and AWS ECS; automated orchestration via Apache Airflow.
- Integrated continuous red teaming tests to verify model robustness and security.

Sample Efficient Rapid Thermal Modeling of 3D IC

- Developed a conditional diffusion model in PyTorch for predicting high-resolution thermal maps from power maps, achieving a 56% performance improvement over traditional GAN models.
- Reduced required training samples to fewer than 800, significantly enhancing training efficiency.
- Optimized model architecture to handle intricate thermal distributions and diverse 3D IC stack layouts, including TSVs and routing congestion effectively.

Predicting Locomotion Modes from Time series Data

- Built a CNN-LSTM hybrid classifier in PyTorch with a 92% F1-score on accelerometer time series, demonstrating expertise in Time Series Analytics and Machine Learning.
- Employed Pandas, NumPy, and Scikit-learn for preprocessing, and implemented cloud-based training on AWS EC2.

Ellipsis — Podcast Generation Platform | Python, Flask, React, Redis, Orpheus TTS

- Developed a fully automated, AI-driven podcast generation agent capable of producing human-like, high-quality podcast on diverse topics and recent news using Perplexity's deep research API.
- Integrated intelligent multi-speaker dialogue generation, significantly enhancing conversational realism and user engagement.